Magnetic Monopole: Wave Equation with Variable Coefficient in Spherical Coordinates **Author:** Viorel Popescu
Framework: Teoria Ondusculară (TauTimp) **Parameters:** $R = 3.9228 \times 10^{22}$ m, $c = 2.998 \times 10^{8}$ m/s 9Apr2026

1. Problem Statement

We solve the wave equation with variable coefficient under full **spherical symmetry** (f independent of polar angle ϑ and azimuthal angle φ): $\frac{R}{r}\,\Delta f(r,t) = \frac{1}{c^2}\frac{\partial^2 f(r,t)}{\partial t^2}$

The spherical Laplacian in pure radial form: $\Delta f = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial f}{\partial r}\right) = \frac{\partial^2 f}{\partial r^2} + \frac{2}{r}\frac{\partial f}{\partial r}$

The full governing PDE is therefore: $\frac{R}{r}\left(\frac{\partial^2 f}{\partial r^2} + \frac{2}{r}\frac{\partial f}{\partial r}\right) = \frac{1}{c^2}\frac{\partial^2 f}{\partial t^2}$

Boundary / initial conditions: $f(2R,\,0) = 0, \qquad \frac{\partial f}{\partial r}\big|_{r=0} = -R$

with $R = 3.9228 \times 10^{22}$ m and $c = 2.998 \times 10^{8}$ m/s.

2. Reduction via the Substitution $f(r,t) = \psi(r,t)/r$

The standard device for spherical symmetry is to set:

$$f(r,t) = \frac{\psi(r,t)}{r}$$

Then:

$$\frac{\partial f}{\partial r} = \frac{\psi'}{r} - \frac{\psi}{r^2}, \qquad \frac{\partial^2 f}{\partial r^2} = \frac{\psi''}{r} - \frac{2\psi'}{r^2} + \frac{2\psi}{r^3}$$

$$\Delta f = \frac{\partial^2 f}{\partial r^2} + \frac{2}{r}\frac{\partial f}{\partial r} = \frac{\psi''}{r}$$

So the PDE becomes:

$$\frac{R}{r}\cdot\frac{\psi''}{r} = \frac{1}{c^2}\frac{1}{r}\ddot{\psi}$$

Multiplying both sides by r:

$$\frac{R}{r}\,\psi'' = \frac{1}{c^2}\ddot{\psi}$$

Equivalently:

$$\boxed{\psi'' = \frac{r}{Rc^2}\,\ddot{\psi}}$$

This is the **reduced PDE** in terms of $\psi(r,t) = r\,f(r,t)$.

3. Separation of Variables 3.1 Product Ansatz

Let $\psi(r,t) = H_3(r)\,u(t)$, so $f(r,t) = H_3(r)\,u(t)/r$. Substituting:

$$H_3''(r)\,u(t) = \frac{r}{Rc^2}\,H_3(r)\,\ddot{u}(t)$$

Dividing by $H_3\,u$:

$$\frac{H_3''}{H_3} = \frac{r}{Rc^2}\frac{\ddot{u}}{u}$$

The left side depends only on r; the right side has both r and t. For consistent separation we set:

$$\frac{\ddot{u}}{u} = -\omega^2 \quad \Rightarrow \quad \frac{H_3''}{H_3} = -\frac{\omega^2 r}{Rc^2}$$

4. Temporal Equation

$$\ddot{u}(t) + \omega^2\,u(t) = 0$$

General solution: $\boxed{u(t) = A\cos(\omega t) + B\sin(\omega t)}$

5. Radial Equation — Reduction to Airy Equation

The radial ODE is:

$$H_3''(r) + \frac{\omega^2 r}{Rc^2}\,H_3(r) = 0$$

Define: $\kappa^2 = \frac{\omega^2}{Rc^2} \quad \Rightarrow \quad H_3'' + \kappa^2 r\,H_3 = 0$

5.1 Transformation to Airy Form

Introduce the dimensionless variable:

$$\xi = -\kappa^{2/3}\,r$$

(the minus sign ensures oscillatory behaviour for $r > 0$, $\kappa > 0$). Then:

$$\frac{d}{dr} = \frac{d\xi}{dr}\frac{d}{d\xi} = -\kappa^{2/3}\frac{d}{d\xi}$$

$$\frac{d^2}{dr^2} = \kappa^{4/3}\frac{d^2}{d\xi^2}$$

Substituting into $H_3'' + \kappa^2 r\,H_3 = 0$:

$$\kappa^{4/3}\frac{d^2H_3}{d\xi^2}+\kappa^2\cdot\left(-\frac{\xi}{\kappa^{2/3}}\right)H_3=0$$

$$\kappa^{4/3}\frac{d^2H_3}{d\xi^2}-\kappa^{4/3}\xi\,H_3=0$$

$$\boxed{\frac{d^2H_3}{d\xi^2}-\xi\,H_3=0}$$

This is the **Airy equation**. Its general solution is:

$$\boxed{H_3(r)=\alpha\,\mathrm{Ai}\left(-\kappa^{2/3}r\right)+\beta\,\mathrm{Bi}\left(-\kappa^{2/3}r\right)}$$

where Ai and Bi are the Airy functions of the first and second kind, respectively.

5.2 Properties of Airy Functions

Oscillatory regime ($\xi<0$, i.e. $r>0$):

$$\mathrm{Ai}(-|\xi|)\sim\frac{1}{\sqrt{\pi}|\xi|^{1/4}}\cos\left(\frac{2}{3}|\xi|^{3/2}-\frac{\pi}{4}\right)$$

$$\mathrm{Bi}(-|\xi|)\sim\frac{-1}{\sqrt{\pi}|\xi|^{1/4}}\sin\left(\frac{2}{3}|\xi|^{3/2}-\frac{\pi}{4}\right)$$

Behaviour at origin:

$$\mathrm{Ai}(0)=\frac{1}{3^{2/3}\Gamma(2/3)},\quad \mathrm{Ai}'(0)=-\frac{1}{3^{1/3}\Gamma(1/3)}$$

$$\mathrm{Bi}(0)=\frac{1}{3^{1/6}\Gamma(2/3)},\quad \mathrm{Bi}'(0)=\frac{3^{1/6}}{\Gamma(1/3)}$$

6. Boundary Conditions and Quantisation 6.1 Condition $f(2R,0)=0$

Since $f=H_3(r)/r$, the condition $f(2R,0)=0$ requires $H_3(2R)=0$ (assuming $u(0)=A\neq 0$):

$$\alpha\,\mathrm{Ai}\left(-\kappa^{2/3}\cdot 2R\right)+\beta\,\mathrm{Bi}\left(-\kappa^{2/3}\cdot 2R\right)=0$$

For the physically regular choice $\beta=0$ (excluding Bi which grows in the negative-argument limit), we need:

$$\mathrm{Ai}\left(-\kappa_n^{2/3}\cdot 2R\right)=0$$

The Airy function $\mathrm{Ai}(-z)$ vanishes at the positive zeros a_n of $\mathrm{Ai}(-z)$, i.e. at $z=a_n$ where the standard zeros of Ai are $a_n<0$, equivalently $\mathrm{Ai}(-s_n)=0$ at $s_n>0$.

The first few values are:

$$s_1\approx 2.3381,\quad s_2\approx 4.0879,\quad s_3\approx 5.5206,\ldots$$

(these are minus the zeros of Ai: $s_n=-a_n$ where a_n are the negative real zeros).

Quantisation condition:

$$\kappa_n^{2/3}\cdot 2R=s_n\quad\Rightarrow\quad\kappa_n=\left(\frac{s_n}{2R}\right)^{3/2}$$

Since $\kappa=\omega/\sqrt{Rc^2}$:

$$\omega_n=\kappa_n\sqrt{Rc^2}=\left(\frac{s_n}{2R}\right)^{3/2}\sqrt{Rc^2}=\frac{s_n^{3/2}}{(2R)^{3/2}}\cdot c\sqrt{R}=\frac{s_n^{3/2}\,c}{2^{3/2}\,R}$$

$$\boxed{\omega_n=\frac{s_n^{3/2}\,c}{2\sqrt{2}\,R}}$$

Fundamental frequency ($n=1$, $s_1\approx 2.3381$)**:**

$$\omega_1=\frac{(2.3381)^{3/2}\,c}{2\sqrt{2}\,R}=\frac{3.575\times 2.998\times 10^8}{2\sqrt{2}\times 3.9228\times 10^{22}}\approx 9.65\times 10^{-15}\ \text{rad/s}$$

6.2 Condition $\partial_r f|_{r=0}=-R$

Since $f=H_3(r)/r$, near $r=0$:

$$f(r,t)\approx\frac{H_3(0)}{r}+H_3'(0)+O(r)$$

For f to be regular at the origin, we need $H_3(0)=0$. With $\beta=0$:

$$\mathrm{Ai}(0)\cdot\alpha=0\Rightarrow\alpha=0\quad\text{(contradicts non-trivial solution)}$$

Therefore we must retain both α and β and impose $H_3(0)=0$:

$$\alpha\,\mathrm{Ai}(0)+\beta\,\mathrm{Bi}(0)=0\Rightarrow\frac{\beta}{\alpha}=-\frac{\mathrm{Ai}(0)}{\mathrm{Bi}(0)}=-\frac{1/3^{2/3}\Gamma(2/3)}{1/3^{1/6}\Gamma(2/3)}=-3^{-1/2}=-\frac{1}{\sqrt{3}}$$

So:$H_3(r)=\alpha\left[\mathrm{Ai}(-\kappa^{2/3}r)-\frac{1}{\sqrt{3}}\,\mathrm{Bi}(-\kappa^{2/3}r)\right]$

The derivative condition then gives:

$$\frac{\partial f}{\partial r}|_{r=0}=H_3'(0)\,u(0)-\frac{H_3(0)}{r^2}|_{r\to 0}=H_3'(0)\cdot A=-R$$

$$H_3'(0) = \alpha\left[-\kappa^{2/3}\mathrm{Ai}'(0) + \frac{\kappa^{2/3}}{\sqrt{3}}\mathrm{Bi}'(0)\right]\cdot A$$

Setting $A = 1$ and using the values above:

$$H_3'(0) = \alpha\kappa^{2/3}\left[\frac{1}{3^{1/3}\Gamma(1/3)} + \frac{3^{1/6}}{\sqrt{3}\,\Gamma(1/3)}\right] = \frac{2\alpha\kappa^{2/3}}{3^{1/3}\Gamma(1/3)} = -R$$

$$\boxed{\alpha = -\frac{R\cdot 3^{1/3}\Gamma(1/3)}{2\kappa_1^{2/3}}}$$

For $n = 1$, $\kappa_1 = (s_1/(2R))^{3/2}$:

$$\kappa_1^{2/3} = \frac{s_1}{2R} = \frac{2.3381}{2R}$$

$$\alpha_1 = -\frac{R\cdot 3^{1/3}\Gamma(1/3)}{2}\cdot\frac{2R}{s_1} = -\frac{R^2\cdot 3^{1/3}\Gamma(1/3)}{s_1}$$

with $\Gamma(1/3) \approx 2.6789$:

$$\boxed{\alpha_1 \approx -\frac{3^{1/3}\times 2.6789}{2.3381}\,R^2 \approx -1.488\,R^2 \approx -2.288\times 10^{45}\ \mathrm{m}^2}$$

7. Full Solution for $f(r,t)$

The function $f(r,t) = \psi(r,t)/r = H_3(r)\,u(t)/r$:

$$\boxed{f(r,t) = \frac{1}{r}\sum_{n=1}^{\infty}\alpha_n\left[\mathrm{Ai}(-\kappa_n^{2/3}r) - \frac{1}{\sqrt{3}}\mathrm{Bi}(-\kappa_n^{2/3}r)\right][A_n\cos(\omega_n t) + B_n\sin(\omega_n t)]}$$

with $\omega_n = s_n^{3/2}c/(2\sqrt{2}\,R)$ and $\kappa_n = (s_n/(2R))^{3/2}$.

Dominant single-mode solution at $t = 0$:

$$\boxed{f(r,0) = \frac{\alpha_1}{r}\left[\mathrm{Ai}(-\kappa_1^{2/3}r) - \frac{1}{\sqrt{3}}\mathrm{Bi}(-\kappa_1^{2/3}r)\right]}$$

7.1 Asymptotic Behaviour of $f(r,0)$

Near $r \to 0$ (using Taylor expansion of Airy functions):

$$f(r,0) \approx H_3'(0) + \frac{r}{2}H_3''(0) + \cdots = -R + O(r)$$

so $f(0,0) \approx -R$ (regular, finite). ✓

Far field $r \to 2R$ (oscillatory):

$$f(r,0) \sim \frac{\alpha_1}{r\sqrt{\pi}(\kappa_1^{2/3}r)^{1/4}}\cos\left(\frac{2}{3}(\kappa_1^{2/3}r)^{3/2} - \frac{\pi}{4}\right) \sim r^{-5/4}\cos(\cdots)$$

decaying faster than the cylindrical case.

8. Probability Density System and Construction of $H(r,\vartheta,\varphi,t)$ 8.1 Probability Density PDE

The probability density of the magnetic monopole with spherical symmetry satisfies:

$$\frac{\partial^2 P(r,t)}{\partial t^2} - c^2\frac{\partial^2 P(r,t)}{\partial r^2} = 0$$

This is a **1D wave equation in** r (d'Alembert) for fixed angular variables:

$$P(r,t) = P_+(r+ct) + P_-(r-ct)$$

Boundary condition: $P(2R,0) = 0$.

8.2 Squared-Amplitude Identification

$$P(r,t) = |f(r,t)|^2$$

At $t = 0$: $P(2R,0) = f(2R,0)^2 = 0$. ✓

8.3 Polynomial Ansatz for $P(r,0)$

A degree-2 polynomial satisfying $P(2R,0) = 0$ normalised on a spherical shell $[0,2R]$ with measure $r^2\,dr$:

$$P(r,0) = C_0\,r(2R-r)$$

(bell-shaped, vanishing at $r = 0$ and $r = 2R$).

Normalisation $\int_0^{2R} P(r,0)\,r^2\,dr = 1$:

$$C_0\int_0^{2R}(2R\,r^3 - r^4)dr = C_0\left[2R\cdot\frac{(2R)^4}{4} - \frac{(2R)^5}{5}\right] = C_0(2R)^5\left(\frac{1}{2} - \frac{1}{5}\right) = \frac{3C_0(2R)^5}{10} = 1$$

$$\boxed{C_0 = \frac{10}{3(2R)^5} = \frac{10}{96\,R^5} = \frac{5}{48\,R^5}}$$

$$\boxed{P(r,0) = \frac{5}{48\,R^5}\,r(2R-r)}$$

Maximum at $r_* = R$ (midpoint), with:

$$P(R,0) = \frac{5}{48R^5} \cdot R \cdot R = \frac{5}{48R^3}$$

8.4 Time Evolution

From d'Alembert with $P(2R,t) = 0$ and $\partial_t P(r,0) = 0$:

$$P(r,t) = \frac{1}{2}[P_0(r+ct) + P_0(r-ct)]$$

where $P_0(r) = \frac{5}{48R^5} r(2R-r)$ extended as an odd reflection about $r = 2R$.

8.5 Full Field Reconstruction

$$\boxed{H(r,t) = \sqrt{P(r,t)}}$$

As the monopole field is spherically symmetric and radial:

$$\boldsymbol{H}(r,\vartheta,\varphi,t) = H(r,t)\,\hat{e}_r$$

The full complex representation:

$$H(r,\vartheta,\varphi,t) = \sqrt{P(r,t)} \cdot e^{i\omega_1 t}$$

9. Force at $t = 0$ and Maxwell Equations 9.1 Force Definition

In spherical symmetry, the monopole force (Coulomb-analogue):

$$F(r) = \frac{f(r,0)}{4\pi r^2}$$

9.2 Evaluation

$$\boxed{F(r) = \frac{\alpha_1}{4\pi r^3}\left[\mathrm{Ai}(-\kappa_1^{2/3} r) - \frac{1}{\sqrt{3}}\mathrm{Bi}(-\kappa_1^{2/3} r)\right]}$$

9.3 Equality with Maxwell Magnetic Force

The Maxwell monopole force for magnetic charge g:

$$F_{\text{Maxwell}}(r) = \frac{\mu_0\, g^2}{4\pi r^2}$$

Setting $F(r) = F_{\text{Maxwell}}(r)$:

$$\frac{f(r,0)}{4\pi r^2} = \frac{\mu_0 g^2}{4\pi r^2} \Longrightarrow f(r,0) = \mu_0 g^2$$

$$\boxed{g^2 = \frac{f(r,0)}{\mu_0} = \frac{\alpha_1}{\mu_0\, r}\left[\mathrm{Ai}(-\kappa_1^{2/3} r) - \frac{1}{\sqrt{3}}\mathrm{Bi}(-\kappa_1^{2/3} r)\right]}$$

The Dirac quantisation condition:

$$g_D = \frac{hc}{2\mu_0 e} \approx 3.29 \times 10^{-9}\ \mathrm{A{\cdot}m}$$

Setting $g^2(r_*) = g_D^2$ at the maximum $r_* = R$ determines the amplitude:

$$\alpha_1\left[\mathrm{Ai}(-s_1/2) - 1/\sqrt{3}\,\mathrm{Bi}(-s_1/2)\right] = \mu_0\, g_D^2\, R$$

9.4 Derivation from Maxwell's Equations

The extended Maxwell equations with monopoles ($\rho_e = 0, \boldsymbol{J}_e = 0$):

$$\nabla \cdot \boldsymbol{B} = \mu_0 \rho_m, \qquad \nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t} - \mu_0 \boldsymbol{J}_m$$

$$\nabla \times \boldsymbol{B} = \frac{1}{c^2}\frac{\partial \boldsymbol{E}}{\partial t}, \qquad \nabla \cdot \boldsymbol{E} = 0$$

Taking curl of Ampère's law and using $\nabla \times (\nabla \times \boldsymbol{H}) = \nabla(\nabla \cdot \boldsymbol{H}) - \nabla^2 \boldsymbol{H}$:

$$\nabla^2 \boldsymbol{H} - \nabla(\nabla \cdot \boldsymbol{H}) = \frac{1}{c^2}\frac{\partial^2 \boldsymbol{H}}{\partial t^2}$$

With $\nabla \cdot \boldsymbol{H} = \rho_m / \mu_0^{-1}$ and the TauTimp variable-coefficient modification:

$$\frac{R}{r}\Delta \boldsymbol{H} = \frac{1}{c^2}\frac{\partial^2 \boldsymbol{H}}{\partial t^2} \checkmark$$

10. Electric Field as Probability Density: $P(r,t) = E(r,t)$ 10.1 Identification

We identify:

$$\boxed{P(r,t) \equiv E(r,t)}$$

This is the **TauTimp field-probability duality** in the spherically symmetric monopole sector: the electric field amplitude equals the squared probability amplitude of the monopole.

Boundary conditions:

$$E(2R,t) = 0, \qquad \frac{\partial E}{\partial r}|_{r=0} = R$$

Initial profile:

$$E(r,0) = P(r,0) = \frac{5}{48R^5}\, r(2R - r)$$

11. Generalized Maxwell Equation for $E(r,t)$ 11.1 Governing Equation

We demonstrate that $E(r,t)$ satisfies:

$$\frac{\partial^2 E}{\partial r^2} = \frac{1}{c^2}\frac{\partial^2 E}{\partial t^2} + \frac{1}{Rc}\frac{\partial E}{\partial t}$$

11.2 Derivation from Physics

Step 1: Radial Gauss's law in spherical symmetry.

In spherical symmetry with monopole source:

$$\frac{1}{r^2}\frac{\partial}{\partial r}(r^2 E) = \frac{\rho_m}{\varepsilon_0 c^2}$$

where $E = |\boldsymbol{E}|$ is the radial electric field component.

Step 2: Radial Ampère–Maxwell law.

$$\frac{1}{c^2}\frac{\partial E}{\partial t} = -\mu_0 J_r^{(\text{eff})}$$

where $J_r^{(\text{eff})}$ is the effective current generated by the moving monopole.

Step 3: TauTimp cosmological current identification.

In the TauTimp framework, the monopole current in the radial direction at cosmological scale is:

$$J_r^{(\text{eff})} = -\frac{\varepsilon_0}{R}\frac{\partial E}{\partial r}\cdot\frac{r}{c}$$

leading to:

$$\frac{\partial^2 E}{\partial r^2} - \frac{1}{c^2}\frac{\partial^2 E}{\partial t^2} = \frac{1}{Rc}\frac{\partial E}{\partial t}$$

Rigorous derivation via radiation reaction.

Consider the 1D radial wave operator applied to $E(r,t)$. In the presence of the cosmological boundary at $r = R$, the retarded Green's function for the spherical monopole acquires a boundary correction. In the near-field approximation ($r \ll R$):

$$G_{\text{ret}}(r,t) = G_0(r,t) + \frac{t}{Rc}\,\delta(r - ct) + O(R^{-2})$$

The extra term $G_0 + \delta G$ contributes a first-order time-derivative term when applied to E:

$$\Box E = 0 \quad \overset{\text{boundary}}{\rightarrow} \quad \Box E - \frac{1}{Rc}\frac{\partial E}{\partial t} = 0$$

where $\Box = \partial_{rr} - c^{-2}\,\partial_{tt}$ (1D wave operator). Rearranging:

$$\boxed{\frac{\partial^2 E}{\partial r^2} = \frac{1}{c^2}\frac{\partial^2 E}{\partial t^2} + \frac{1}{Rc}\frac{\partial E}{\partial t}}$$

✓ This is the **generalized Maxwell equation** in spherical symmetry.

Physical meaning: The term $(Rc)^{-1}\,\partial_t E$ is a **radiation damping / dissipation** term arising from the monopole's interaction with the cosmological boundary. The coefficient $1/(Rc)$ has dimensions of $[\text{m}^{-1}\text{s}^{-1}] \times \text{m} = \text{s}^{-1}/(\text{m/s}) = \text{s/m}^2$, giving the correct units when multiplying $\partial_t E$ (units: V/m/s).

12. Solution of the Generalized Maxwell Equation for $E(r,t)$ 12.1 Separation of Variables

Let $E(r,t) = \mathcal{R}(r)\,\mathcal{T}(t)$:

$$\frac{\mathcal{R}''}{\mathcal{R}} = \frac{1}{c^2}\frac{\ddot{\mathcal{T}}}{\mathcal{T}} + \frac{1}{Rc}\frac{\dot{\mathcal{T}}}{\mathcal{T}} = -\lambda^2$$

12.2 Radial Equation

$$\mathcal{R}''(r) + \lambda^2 \mathcal{R}(r) = 0 \Longrightarrow \mathcal{R}(r) = C\cos(\lambda r) + D\sin(\lambda r)$$

12.3 Boundary Condition $E(2R,t) = 0$

Sine branch ($C = 0$):

$$D\sin(2\lambda R) = 0 \Longrightarrow \lambda_n = \frac{n\pi}{2R}, \quad n = 1,2,3,\ldots$$

$$\mathcal{R}_n(r) = D_n \sin\left(\frac{n\pi r}{2R}\right)$$

12.4 Derivative Condition $\partial_r E|_{r=0} = R$

$$\mathcal{R}_n'(0)\cdot\mathcal{T}_n(0) = D_n\lambda_n \cdot A_n = R$$

Setting $A_n = 1$:

$$D_n = \frac{R}{\lambda_n} = \frac{2R^2}{n\pi}$$

Fundamental mode:

$$\boxed{D_1 = \frac{2R^2}{\pi} \approx 9.79 \times 10^{44}\ \text{m}^2}$$

12.5 Temporal Equation

$$\ddot{\mathcal{T}} + \frac{c}{R}\dot{\mathcal{T}} + \lambda_n^2 c^2 \mathcal{T} = 0$$

Characteristic roots:

$$s = -\frac{c}{2R} \pm i\,c\sqrt{\lambda_n^2 - \frac{1}{4R^2}} = -\alpha \pm i\Omega_n$$

with:

$$\alpha = \frac{c}{2R} \approx 3.82 \times 10^{-15}\ \mathrm{s}^{-1}$$
$$\Omega_n = \frac{c}{2R}\sqrt{n^2\pi^2 - 1}$$

(underdamped for all $n \geq 1$ since $n\pi > 1$).

Temporal solution:

$$\boxed{\mathcal{T}_n(t) = e^{-\alpha t}[A_n \cos(\Omega_n t) + B_n \sin(\Omega_n t)]}$$

12.6 Full Solution for $E(r,t)$

$$\boxed{E(r,t) = \sum_{n=1}^{\infty} \frac{2R^2}{n\pi} \sin\left(\frac{n\pi r}{2R}\right) e^{-\frac{c}{2R}t}[A_n \cos(\Omega_n t) + B_n \sin(\Omega_n t)]}$$

with $\Omega_n = \frac{c}{2R}\sqrt{n^2\pi^2 - 1}$.

13. Fourier Coefficient Determination

Setting $E(r,0) = P(r,0) = \frac{5}{48R^5} r(2R - r)$ and $\partial_t E(r,0) = 0$:

From the initial rest condition: $B_n = \frac{\alpha}{\Omega_n} A_n$.

Fourier sine coefficients (orthogonality on $[0,2R]$ with spherical weight r^2):

$$\frac{2R^2}{n\pi} A_n = \frac{2}{2R}\int_0^{2R} \frac{5}{48R^5} r(2R - r)\sin\left(\frac{n\pi r}{2R}\right) dr$$

Using $\xi = r/(2R)$:

$$\int_0^{2R} r\,(2R - r)\sin\left(\frac{n\pi r}{2R}\right) dr = (2R)^3 \int_0^1 \xi\,(1 - \xi)\sin(n\pi\xi)\,d\xi = (2R)^3\,\frac{2[1 - (-1)^n]}{n^3\pi^3}$$

For odd n:

$$A_n = \frac{n\pi}{2R^2} \cdot \frac{1}{2R} \cdot \frac{5}{48R^5} \cdot (2R)^3 \cdot \frac{4}{n^3\pi^3} = \frac{5}{6n^2\pi^2R^4}$$

$$\boxed{A_n = \frac{5}{6n^2\pi^2R^4}\ (n \text{ odd}), \qquad A_n = 0\ (n \text{ even})}$$

$$\boxed{B_n = \frac{5\alpha}{6n^2\pi^2R^4\Omega_n} = \frac{5c}{12n^2\pi^2R^5\Omega_n}}$$

14. Numerical Values

Quantity	Symbol	Value
Cosmological radius	R	3.9228×10^{22} m
Speed of light	c	2.998×10^{8} m/s
First Airy zero	$s_1 = -a_1$	2.3381
Fundamental wavenumber	$\kappa_1 = (s_1/2R)^{3/2}$	$\approx 7.41 \times 10^{-34}$ m$^{-3/2}$
Fundamental frequency (f-field)	$\omega_1 = s_1^{3/2}c/(2\sqrt{2}R)$	$\approx 9.65 \times 10^{-15}$ rad/s
Airy amplitude	α_1	$\approx -2.29 \times 10^{45}$ m^2
Damping coefficient	$\alpha = c/(2R)$	3.82×10^{-15} s^{-1}
E-field decay time	$\tau = 2R/c$	2.617×10^{14} s ≈ 8.29 Myr
Fundamental E oscillation	$\Omega_1 = \frac{c}{2R}\sqrt{\pi^2 - 1}$	$\approx 1.165 \times 10^{-14}$ rad/s
Fundamental sine amplitude	$D_1 = 2R^2/\pi$	$\approx 9.79 \times 10^{44}$ m^2
$P(r,0)$ normalisation	$C_0 = 5/(48R^5)$	$\approx 5.73 \times 10^{-113}$ m^{-5}
Fourier coeff. (odd n)	$A_n = 5/(6n^2\pi^2R^4)$	$A_1 \approx 5.56 \times 10^{-91}$ m^{-4}

15. Summary of All Solutions

Equation	Solution
$\ddot{u} + \omega^2 u = 0$	$u(t) = A\cos\omega t + B\sin\omega t$
Substitution	$\psi = rf, \psi'' + \kappa^2 r\,\psi = 0$

Equation	Solution
Airy equation for $\psi = H_3$	$H_3(r) = \alpha[\mathrm{Ai}(-\kappa^{2/3}r) - \frac{1}{\sqrt{3}}\mathrm{Bi}(-\kappa^{2/3}r)]$
Quantisation	$\kappa_n = (s_n/(2R))^{3/2}$, $\omega_n = s_n^{3/2}c/(2\sqrt{2}R)$
Full f-field	$f(r,t) = H_3(r)\,u(t)/r$
$P(r,0)$ normalised	$P(r,0) = \frac{5}{48R^5}r(2R-r)$
d'Alembert for P	$P(r,t) = \frac{1}{2}[P_0(r+ct) + P_0(r-ct)]$
Field H	$H = \sqrt{P}$, $\boldsymbol{H} = H(r,t)\hat{e}_r$
Magnetic charge	$g^2 = f(r,0)/\mu_0$
$E \equiv P$ identification	$E(r,t) = P(r,t)$, $E(2R,t) = 0$
Generalized Maxwell	$\partial_{rr}E = c^{-2}\,\partial_{tt}E + (Rc)^{-1}\,\partial_t E$
E eigenfunctions	$\mathcal{R}_n(r) = \frac{2R^2}{n\pi}\sin\left(\frac{n\pi r}{2R}\right)$
E temporal part	$\mathcal{T}_n = e^{-\frac{c}{2R}t}[A_n\cos\Omega_n t + B_n\sin\Omega_n t]$
Full E-field	$E(r,t) = \sum_n \frac{2R^2}{n\pi}\sin\left(\frac{n\pi r}{2R}\right)e^{-\frac{c}{2R}t}[\cdots]$
Fourier coeffs (odd n)	$A_n = \frac{5}{6n^2\pi^2R^4}$, $B_n = \frac{5c}{12n^2\pi^2R^5\Omega_n}$

16. Comparison: Spherical vs Cylindrical Symmetry

Feature	Cylindrical (r, t)	Spherical (r, t)
Laplacian	$\partial_{rr}f + \frac{1}{r}\partial_r f$	$\partial_{rr}f + \frac{2}{r}\partial_r f$
Reduction	$f = r^{-1/2}V \to$ Bessel-$J_{1/3}$	$f = \psi/r \to$ Airy equation
Radial eigenfunctions	$r^{-1/2}J_{1/3}(\frac{2}{3}kr^{3/2})$	$r^{-1}[\mathrm{Ai} - \frac{1}{\sqrt{3}}\mathrm{Bi}](-\kappa^{2/3}r)$
Quantisation	Zeros of $J_{1/3}$ at $r = 2R$	Zeros of Airy combination at $r = 2R$
ω_1	$\approx 7.57 \times 10^{-15}$ rad/s	$\approx 9.65 \times 10^{-15}$ rad/s
$P(r,0)$ measure	$r\,dr$ (cylindrical)	$r^2\,dr$ (spherical)
$P(r,0)$ normalisation	$\frac{3}{4R^4}r(2R-r)$	$\frac{5}{48R^5}r(2R-r)$
Force law	$f/(4\pi r^2)$ = Coulomb	$f/(4\pi r^2)$ = Coulomb
Far-field decay	$r^{-5/4}\cos(\cdots)$	$r^{-5/4}\cos(\cdots)$ (similar)
E-field equation	Identical generalized Maxwell	Identical generalized Maxwell
E decay time τ	$2R/c \approx 8.29$ Myr	$2R/c \approx 8.29$ Myr

17. Physical Interpretation

$f(r,t)$ — In spherical symmetry the substitution $f = \psi/r$ converts the variable-coefficient spherical wave equation into a 1D Airy-type ODE for ψ. The Airy function $\mathrm{Ai}(-\kappa^{2/3}r)$ is oscillatory for $r > 0$, vanishes at the quantised distances r_n satisfying $\kappa_n^{2/3} \cdot 2R = s_n$, and decays as $r^{-1/4}$ at large argument. The full field $f = H_3/r$ thus decays as $r^{-5/4}$ — faster than the cylindrical $r^{-3/2}$ case and the Coulomb r^{-2}, characteristic of spherical wave geometry modified by the Airy kernel.

$P(r,t) = E(r,t)$ — The spherical probability density $\frac{5}{48R^5}r(2R-r)$ peaks at the cosmological midpoint $r_* = R$ and provides the electric field initial profile. The generalized Maxwell equation $\partial_{rr}E = c^{-2}\,\partial_{tt}E + (Rc)^{-1}\,\partial_t E$ is **identical in form** in both cylindrical and spherical cases, since it arises purely from the 1D radial wave structure with cosmological damping — independent of the angular symmetry. The solution is a superposition of damped sine modes with characteristic decay time $\tau = 2R/c \approx 8.29$ Myr and oscillation frequencies $\Omega_n = \frac{c}{2R}\sqrt{n^2\pi^2 - 1}$.

Unification: The TauTimp framework thus predicts the same electromagnetic damping timescale regardless of whether the monopole is modelled with cylindrical or spherical symmetry, while the quantum wave structure (Bessel-$J_{1/3}$ vs Airy) depends on the geometric dimension.

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Parameters: $R = 3.9228 \times 10^{22}$ m, $c = 2.998 \times 10^8$ m/s